

Thm (invariance of dimension)

Let V be a finite dimensional vector space (i.e. V has a finite spanning set). Then any two bases of V have the same (necessarily finite) cardinality

Corollaries Let $\dim V = n$. Then

- (b) any spanning set S with $|S| = n$, is automatically lin. indep & hence a basis (pf Use strategy B to drop vectors from S to get a basis. As any basis has size n , S itself must have been a basis.)
- (a) any lin indep set I with $|I| = n$ automatically spans V and hence is a basis. (Use strategy A ... parallel to above)
- (c) Suppose W is a subspace of V . Then $\dim W \leq \dim V$ with equality if and only if $W = V$
pf. Take a basis of W . It stays lin indep as a subset of V .
Extend to get a basis of V using strategy A. If $W \neq V$ then I have to extend (This gives "only if")
because the given basis spans on W , so I have to add to it to get V .

Proof of thm will be based on the following proposition.

Proposition:
Suppose $\begin{cases} V \text{ is any vector space.} \\ I = \{v_1, \dots, v_n\} \text{ a lin indep set in } V \\ S \in V \text{ s.t. } \text{span}(S) = V \end{cases}$

Then $|I| = n \leq |S|$

Pf. If S is infinite there is nothing to prove, so let $S = \{v_1, \dots, v_k\}$. Want to show $k \geq n$.

We will insert $l_1, l_2, \dots$ into S one at a time maintaining span and size of S . How? Let $S_0 = S$

$l_1 \in \text{Span}(S)$. So write $l_1 = a_1 v_1 + a_2 v_2 + \dots + a_k v_k$

Not all $a_i = 0$ (b/c $l_1 \neq 0$). WLOG let $a_1 \neq 0$. Replace

v_1 by l_1 in S to get S_1 i.e. $S_1 = S_0 - \{v_1\} \cup \{l_1\}$.

$\text{Span}(S_1) = \text{Span}(S)$ { b/c $l_1 \in \text{Span}(S_0)$ and $v_1 \in \text{Span}(S_1)$
and other elements are same in S_0 and S_1

Continue: From $S_i = \{l_1, l_2, \dots, l_i, v_{i+1}, \dots, v_k\}$ (by renumbering v 's as necessary)
get $S_{i+1} = \{l_1, l_2, \dots, l_i, l_{i+1}, v_{i+1}, \dots, v_k\}$

We can do this till we run out of l 's or v 's.

Two cases: If $n \leq k$ we are done

If $n > k$ then we stop with $S_k = \{l_1, \dots, l_k\}$, so (at least) $l_{k+1} \notin S_k$. But $\text{Span}(S_k) = V$ so

$l_{k+1} = \text{lin comb of } l_1, \dots, l_k$. This contradicts lin indep. of I . This proves the proposition

More formally we could have said: Consider all subsets T of $S \cup I$ with the following properties: $\begin{cases} |T| \leq |S| \\ \text{Span}(T) = V \end{cases}$

Among these pick one with maximum possible $|T \cap I|$.

Try to complete the pf with this setup.

proof of the thm Suppose B_1 & B_2 are bases of f.d. V .

We will see ^(in the note below) that B_1, B_2 must be finite. So apply proposition:

As B_1 is indep & B_2 spans $|B_1| \leq |B_2|$
As B_2 is indep & B_1 spans $|B_2| \leq |B_1|$ } $\Rightarrow |B_1| = |B_2|$

Note Any linearly indep set in a f.d. n V space must be finite
We need to prove this without assuming the theorem

But this is easy b/c any subset of indep set stays indep.

and by proposition no finite indep set can have cardinality greater than any spanning set S and we have a finite S.

For vector spaces that are not f.dimensional **one can**

- show existence of basis using strategy A and Zorn's lemma
- uniqueness of dimension uses some set theory.

How to calculate (with) bases

ordered
1) Take 2 basis $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \end{bmatrix}$ of $\mathbb{R}^2$.

Exercise Find coordinates of $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$ wrt the given basis. i.e

Find a, b scalars s.t. $a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 3 \\ 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$

$$\Leftrightarrow \begin{bmatrix} a+3b \\ 2a+8b \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 1 & 3 \\ 2 & 8 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} \quad \text{Solve!}$$

2) $V = \{ \text{poly of deg} \leq 2 \}$. Take basis $1, 1+x, 1+x+x^2$

Observe that this is lin indep! Basis b/c $\dim V = 3$.

$$a + b(1+x) + c(1+x+x^2) = 0 \Rightarrow \text{first } c=0, \text{ then } b=0 \text{ then } a=0$$

Exercise Find coeffs of $3+4x-9x^2$ wrt this given basis

3) Take $r \times c$ matrix A , giving map $f_A: \mathbb{R}^c \rightarrow \mathbb{R}^r$
 $x \mapsto Ax$

How to find basis of (i) $\ker f_A = \text{nul}(A)$

(ii) $\text{im } f_A = \{Ax \mid x \in \mathbb{R}^c\} = \text{span}(\text{cols of } A)$
 $= \text{column space of } A$

(i) $\text{Nul}(A) = \{x \mid Ax=0\} = \{x \mid Rx=0\}$ where

$A \xrightarrow[\text{reduce}]{\text{row}} R = EA$ We'll see an example

$$\begin{bmatrix} 0 & 1 & p & 0 & q & 0 \\ 0 & 0 & 0 & 1 & r & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

free variable columns

Soln $x_1 = x_1$

$$x_2 = -px_3 - qx_5$$

$$x_3 = x_3$$

$$x_4 = -rx_5$$

$$x_5 = x_5$$

$$x_6 = 0$$

i.e. Any solution $\bar{x}$ is of form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -p \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ -q \\ 0 \\ -r \\ 1 \\ 0 \end{bmatrix}$$

These three vectors give a basis for nul space. (dim = 3)

[They span, from work above. Why lin indep? Look at slots 1, 3, 5]

Can do this in general!

Conclusion $\dim \text{Nul space} = \# \text{ of free variables}$

- (ii) Solve this & see $\dim \text{Col space} = ?$ } Exercise.
(iii) How to find basis (& $\dim$) of row space ? }

It is extremely worthwhile to work out a good example
and justify all steps using the theory.